

Multi-dimensional continuous time quantum walks related to the birth and death chains

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In this paper, we consider multi-dimensional birth and death chains and continuous time quantum walks (CTQWs) related to them. For continuous time quantum walk (CTQW) related to our forms of multi-dimensional birth and death chains, we obtain the time-scaled independence between multiple dimensions about the transition probability of CTQW. By using this feature, we analyze CTQW on the path graph, which is related to 1-dimensional Ehrenfest model. We also have a random variable which is related to our models and converges to the standard Gaussian distribution.

Keywords: Quantum walks; continuous time; birth and death chains.

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1. Introduction

During recent two decades, the quantum walks (QWs) have been intensively studied^{1–9} because of the various applicable fields. QWs can be regarded as the quantum counterpart of the classical random walks (RWs). QWs have several different features from those of RWs.^{10,11}

On the other hand, the 1-dimensional Ehrenfest model is widely studied in many fields because this simple model describes many random motions and phenomena.¹² It is known that the Ehrenfest model can be described and analyzed by the tensor product of RWs on the simple edge. In addition, the more general models, called the birth and death chains, are also studied in many fields.^{13,14} In the previous study,¹⁵ the discrete time quantum walks on the path graph, which are related to the birth and death chains, are analyzed.

In this paper, we consider the multi-dimensional versions of the Ehrenfest model and birth and death chains, which are more general models compared with the Ehrenfest model, and analyze the continuous time quantum walk (CTQW) related to them. The procedure of the multi-dimensional birth and death chains is to select the dimension and increase or decrease the value of the coordinate of the selected dimension. Continuous time quantum walk (CTQW) related to the models is regarded as QWs with multi walkers. Then we obtain that the transition probability of the CTQW can be written by the joint probability distribution of independent quantum walkers, whose time is scaled by the select probability of the each walker.

This paper is arranged as follows. In Sec. 2, we present the definition of our models. Section 2.1 deals with the 2-dimensional Ehrenfest model as an example, and after that we consider multi-dimensional birth and death chains. In Sec. 2.2, we define CTQW related to the models given in Sec. 2.1. Section 3 is devoted to our main results about CTQW.

2. Definition

In this section, we define the multi-dimensional Ehrenfest model, birth and death chains and CTQW on the path.

2.1. The multi-dimensional models

In this subsection, we focus on the definition the d -dimensional Ehrenfest model ($d = 1, 2, \dots$).

2.1.1. 2-dimensional Ehrenfest model

At first, we discuss the 2-dimensional Ehrenfest model as an example. We consider a total of N balls, divided into 2 kinds, contained in two boxes, A and B . Let $N^{(1)}$ be the number of the first kind of balls and $N^{(2)}$ be the number of the second kind of balls in boxes A and B , where $N^{(1)} + N^{(2)} = N$. We put $X_t^{(i)}$ ($i = 1, 2$) as the

random variable on $\mathbb{Z}$, which denotes the number of i th kind of balls in box A at time $t \in \mathbb{Z}_{\geq 0}$. Then the transition probability of $X_t^{(i)}$ is written as follows:

$$\left\{ \begin{array}{ll} \mathbb{P}(X_{t+1}^{(i)} = k \mid X_t^{(i)} = k) = \frac{N - N^{(i)}}{N} = 1 - \frac{N^{(i)}}{N} & (k = 0, \dots, N^{(i)}), \\ \mathbb{P}(X_{t+1}^{(i)} = k + 1 \mid X_t^{(i)} = k) = \frac{N^{(i)} - k}{N} \\ \quad = \frac{N^{(i)}}{N} \left(1 - \frac{k}{N^{(i)}} \right) & (k = 0, \dots, N^{(i)} - 1), \\ \mathbb{P}(X_{t+1}^{(i)} = k - 1 \mid X_t^{(i)} = k) = \frac{k}{N} = \frac{N^{(i)}}{N} \left(\frac{k}{N^{(i)}} \right) & (k = 1, \dots, N^{(i)}), \\ \mathbb{P}(X_{t+1}^{(i)} = j \mid X_t^{(i)} = k) = 0 & (\text{otherwise}). \end{array} \right.$$

These equations mean that a ball is selected at random and moved to the other box. Let P_N be the transition probability matrix of the Ehrenfest model written by

$$P_N = \frac{N^{(1)}}{N} (P_{N^{(1)}} \otimes I_{N^{(2)}}) + \frac{N^{(2)}}{N} (I_{N^{(1)}} \otimes P_{N^{(2)}}),$$

where $P_{N^{(i)}} (i = 1, 2)$ are the transition probability matrices whose k, l elements are $\mathbb{P}(X_{t+1}^{(i)} = l \mid X_t^{(i)} = k)$. We define the distribution p_t as follows:

$$p_t(x^{(1)}, x^{(2)}) = \mathbb{P}(X_t^{(1)} = x^{(1)}, X_t^{(2)} = x^{(2)}).$$

Then we put the operator $M_t : \mathbb{Z} \otimes \mathbb{Z} \rightarrow [0, 1]$ whose operation is written by

$$M_t = \sum_{0 \leq x^{(1)} \leq N^{(1)}} \sum_{0 \leq x^{(2)} \leq N^{(2)}} \{p_t(x^{(1)}, x^{(2)}) \langle x^{(1)} | \otimes \langle x^{(2)} | \}.$$

We should note that $M_{t+1} = M_t P_N$ and

$$M_t = M_0 P_N^t.$$

Next, let $\{\pi^{(i)}(j)\}$ ($i = 1, 2, j = 0, \dots, N^{(i)}$) be the stationary distribution which satisfies the detailed balance condition of the Ehrenfest model.

We define the matrix

$$J_{N^{(i)}} = D_{N^{(i)}}^{1/2} P_{N^{(i)}} D_{N^{(i)}}^{-1/2},$$

where $D_{N^{(i)}} = \text{diag}(\pi^{(i)}(0), \dots, \pi^{(i)}(N^{(i)}))$. Since each of $J_{N^{(i)}}$ is a real symmetric matrix, we have the following spectral decomposition:

$$J_{N^{(i)}} = \sum_{l^{(i)}=0}^{N^{(i)}} \lambda_{l^{(i)}}^{(i)} |\mathbf{v}_{l^{(i)}}^{(i)}\rangle \langle \mathbf{v}_{l^{(i)}}^{(i)}|, \quad (1)$$

where $\lambda_{l^{(i)}}^{(i)}$ is the $l^{(i)}$ th eigenvalue of $J_{N^{(i)}}$ and

$$|\mathbf{v}_{l^{(i)}}^{(i)}\rangle = \begin{bmatrix} p_{l^{(i)}}^{(i)}(0) \\ p_{l^{(i)}}^{(i)}(1) \\ \vdots \\ p_{l^{(i)}}^{(i)}(N^{(i)}) \end{bmatrix} \quad v_{l^{(i)}}^{(i)} = \begin{bmatrix} 1 \\ p_{l^{(i)}}^{(i)}(1) \\ \vdots \\ p_{l^{(i)}}^{(i)}(N^{(i)}) \end{bmatrix} \quad v_{l^{(i)}}^{(i)}$$

is the corresponding eigenvector with $v_{l^{(i)}}^{(i)}, p_{l^{(i)}}^{(i)}(j) \in \mathbb{R}, (j = 0, \dots, N^{(i)})$. Noting that $p_{l^{(i)}}^{(i)}$ can be regarded as function of $\lambda_{l^{(i)}}^{(i)}$, we define the measure function $\mu^{(i)}$ by

$$\mu^{(i)}(\lambda_{l^{(i)}}^{(i)}) = |v_{l^{(i)}}^{(i)}|^2. \quad (2)$$

We put the matrix J_N as follows:

$$J_N = \frac{N^{(1)}}{N}(J_{N^{(1)}} \otimes I_{N^{(2)}}) + \frac{N^{(2)}}{N}(I_{N^{(1)}} \otimes J_{N^{(2)}}).$$

By Eq. (1), we obtain

$$J_N(|\mathbf{v}_{l^{(1)}}^{(1)}\rangle \otimes |\mathbf{v}_{l^{(2)}}^{(2)}\rangle) = \left(\frac{N^{(1)}}{N}\lambda_{l^{(1)}}^{(1)} + \frac{N^{(2)}}{N}\lambda_{l^{(2)}}^{(2)} \right) (|\mathbf{v}_{l^{(1)}}^{(1)}\rangle \otimes |\mathbf{v}_{l^{(2)}}^{(2)}\rangle).$$

Let V be the orthogonal matrix written by

$$\begin{aligned} V = [& |v_0^{(1)}\rangle \otimes |v_0^{(2)}\rangle, \quad |v_0^{(1)}\rangle \otimes |v_1^{(2)}\rangle, \quad \dots \quad |v_0^{(1)}\rangle \otimes |v_{N^{(2)}}^{(2)}\rangle, \\ & |v_1^{(1)}\rangle \otimes |v_0^{(2)}\rangle, \quad |v_1^{(1)}\rangle \otimes |v_1^{(2)}\rangle, \quad \dots \quad |v_1^{(1)}\rangle \otimes |v_{N^{(2)}}^{(2)}\rangle, \\ & \dots \quad |v_{N^{(1)}}^{(1)}\rangle \otimes |v_{N^{(2)}}^{(2)}\rangle]. \end{aligned}$$

Since V and *V are orthogonal matrices, we have the following another description of $V^*V = I$:

$$\begin{aligned} \delta_{\mathbf{j},\mathbf{k}} &= \sum_{l^{(1)}=0}^{N^{(1)}} \sum_{l^{(2)}=0}^{N^{(2)}} \left(\overline{v_{l^{(1)}}^{(1)} p_{l^{(1)}}^{(1)}(j^{(1)}) v_{l^{(2)}}^{(2)} p_{l^{(2)}}^{(2)}(j^{(2)})} v_{l^{(1)}}^{(1)} p_{l^{(1)}}^{(1)}(k^{(1)}) v_{l^{(2)}}^{(2)} p_{l^{(2)}}^{(2)}(k^{(2)}) \right) \\ &= \sum_{l^{(1)}=0}^{N^{(1)}} \sum_{l^{(2)}=0}^{N^{(2)}} \left(\overline{p_{l^{(1)}}^{(1)}(j^{(1)}) p_{l^{(2)}}^{(2)}(j^{(2)})} p_{l^{(1)}}^{(1)}(k^{(1)}) p_{l^{(2)}}^{(2)}(k^{(2)}) |v_{l^{(1)}}^{(1)}|^2 |v_{l^{(2)}}^{(2)}|^2 \right) \\ &= \sum_{l^{(1)}=0}^{N^{(1)}} \sum_{l^{(2)}=0}^{N^{(2)}} \left(\overline{p_{l^{(1)}}^{(1)}(j^{(1)}) p_{l^{(2)}}^{(2)}(j^{(2)})} p_{l^{(1)}}^{(1)}(k^{(1)}) p_{l^{(2)}}^{(2)}(k^{(2)}) \mu^{(1)}(\lambda_{l^{(1)}}^{(1)}) \mu^{(2)}(\lambda_{l^{(2)}}^{(2)}) \right), \end{aligned}$$

where $\mathbf{j} = [j^{(1)}, j^{(2)}]$, $\mathbf{k} = [k^{(1)}, k^{(2)}]$ and

$$\delta_{\mathbf{j},\mathbf{k}} = \begin{cases} 1 & (\mathbf{j} = \mathbf{k}), \\ 0 & (\mathbf{j} \neq \mathbf{k}). \end{cases}$$

2.1.2. Multi-dimensional birth and death chains

Next, we consider more general and higher dimensional models, so-called birth and death chains in the same manner. We consider RW on the d -dimensional tensor product $N^{(1)} \otimes \cdots \otimes N^{(d)}$ where $N^{(1)} + \cdots + N^{(d)} = N$. i th coordinate ($i = 1, \dots, d$) means the number of living i th creature in birth and death chains. Let $q^{(i)}$ be the selection probability of i th dimension among d dimensions and $p^{(i)}$ be the probability function that the value of i th coordinate decreases by 1. We put $X_t^{(i)}$ ($i = 1, \dots, d$) as the random variable on $\mathbb{Z}$, which denotes the value of i th coordinate at time $t \in \mathbb{Z}_{\geq 0}$. Then the transition probability of $X_t^{(i)}$ is written as follows:

$$\begin{cases} \mathbb{P}(X_{t+1}^{(i)} = k \mid X_t^{(i)} = k) = 1 - q^{(i)} & (k = 0, \dots, N^{(i)}), \\ \mathbb{P}(X_{t+1}^{(i)} = 1 \mid X_t^{(i)} = 0) = q^{(i)} \times 1, \\ \mathbb{P}(X_{t+1}^{(i)} = k + 1 \mid X_t^{(i)} = k) = q^{(i)}(1 - p^{(i)}(k)) & (k = 1, \dots, N^{(i)} - 1), \\ \mathbb{P}(X_{t+1}^{(i)} = k - 1 \mid X_t^{(i)} = k) = q^{(i)}p^{(i)}(k) & (k = 1, \dots, N^{(i)} - 1), \\ \mathbb{P}(X_{t+1}^{(i)} = N^{(i)} - 1 \mid X_t^{(i)} = N^{(i)}) = q^{(i)} \times 1, \\ \mathbb{P}(X_{t+1}^{(i)} = j \mid X_t^{(i)} = k) = 0 & (\text{otherwise}). \end{cases}$$

For simplicity, we assume $0 < p^{(i)}(k) < 1$ ($k = 1, \dots, N^{(i)} - 1$). $\mathbb{P}(X_{t+1}^{(i)} = 1 \mid X_t^{(i)} = 0) = q^{(i)}$ and $\mathbb{P}(X_{t+1}^{(i)} = N^{(i)} - 1 \mid X_t^{(i)} = N^{(i)}) = q^{(i)}$ means that $p^{(i)}(0) = 0$, $p^{(i)}(N^{(i)}) = 1$, in other words, we put the reflective boundary conditions for each dimension. Let P_N be the transition probability matrix of birth and death chains written by

$$P_N = q^{(1)}(P_{N^{(1)}} \otimes I_{N^{(2)}} \otimes \cdots \otimes I_{N^{(d)}}) + \cdots + q^{(d)}(I_{N^{(1)}} \otimes I_{N^{(2)}} \otimes \cdots \otimes P_{N^{(d)}}),$$

where $P_{N^{(i)}} (i = 1, \dots, d)$ are the transition probability matrices whose k, l elements are $\mathbb{P}(X_{t+1}^{(i)} = l \mid X_t^{(i)} = k)$. We define the distribution p_t as follows:

$$p_t(x^{(1)}, \dots, x^{(d)}) = \mathbb{P}(X_t^{(1)} = x^{(1)}, \dots, X_t^{(d)} = x^{(d)}).$$

Then we put the operator $M_t : \mathbb{Z}^{\otimes d} \rightarrow [0, 1]$ whose operation is written by

$$M_t = \sum_{0 \leq x^{(1)} \leq N^{(1)}} \cdots \sum_{0 \leq x^{(d)} \leq N^{(d)}} \{p_t(x^{(1)}, \dots, x^{(d)}) \langle x^{(1)} \mid \otimes \cdots \otimes \langle x^{(d)} \mid \}.$$

We should note that $M_{t+1} = M_t P_N$ and

$$M_t = M_0 P_N^t.$$

Next, let $\{\pi^{(i)}(j)\}$ ($i = 1, \dots, d$) and ($j = 0, \dots, N^{(i)}$) be the stationary distribution which satisfies the detailed balance condition of birth and death chains.

We define the matrix

$$J_{N^{(i)}} = D_{N^{(i)}}^{1/2} P_{N^{(i)}} D_{N^{(i)}}^{-1/2},$$

where $D_{N^{(i)}} = \text{diag}(\pi^{(i)}(0), \dots, \pi^{(i)}(N^{(i)}))$. Since each of $J_{N^{(i)}}$ is a real symmetric matrix, we have the following spectral decomposition:

$$J_{N^{(i)}} = \sum_{l^{(i)}=0}^{N^{(i)}} \lambda_{l^{(i)}}^{(i)} |\mathbf{v}_{l^{(i)}}^{(i)}\rangle \langle \mathbf{v}_{l^{(i)}}^{(i)}|, \quad (3)$$

where $\lambda_{l^{(i)}}^{(i)}$ is the $l^{(i)}$ th eigenvalue of $J_{N^{(i)}}$ and

$$|\mathbf{v}_{l^{(i)}}^{(i)}\rangle = \begin{bmatrix} p_{l^{(i)}}^{(i)}(0) \\ p_{l^{(i)}}^{(i)}(1) \\ \vdots \\ p_{l^{(i)}}^{(i)}(N^{(i)}) \end{bmatrix} \quad v_{l^{(i)}}^{(i)} = \begin{bmatrix} 1 \\ p_{l^{(i)}}^{(i)}(1) \\ \vdots \\ p_{l^{(i)}}^{(i)}(N^{(i)}) \end{bmatrix} \quad v_{l^{(i)}}^{(i)}$$

is the corresponding eigenvector with $v_{l^{(i)}}^{(i)}, p_{l^{(i)}}^{(i)}(j) \in \mathbb{R}, (j = 0, \dots, N^{(i)})$. Owing to $p_{l^{(i)}}^{(i)}$ can be regarded as function of $\lambda_{l^{(i)}}^{(i)}$, we define the measure function $\mu^{(i)}$ by

$$\mu^{(i)}(\lambda_{l^{(i)}}^{(i)}) = |v_{l^{(i)}}^{(i)}|^2. \quad (4)$$

We put the matrix J_N as follows:

$$J_N = q^{(1)}(J_{N^{(1)}} \otimes I_{N^{(2)}} \otimes \dots \otimes I_{N^{(d)}}) + \dots + q^{(d)}(I_{N^{(1)}} \otimes I_{N^{(2)}} \otimes \dots \otimes J_{N^{(d)}}).$$

By Eq. (3), we obtain

$$J_N \left(\bigotimes_{i=1}^d |\mathbf{v}_{l^{(i)}}^{(i)}\rangle \right) = \left(\sum_{i=1}^d q^{(i)} \lambda_{l^{(i)}}^{(i)} \right) \left(\bigotimes_{i=1}^d |\mathbf{v}_{l^{(i)}}^{(i)}\rangle \right).$$

Let V_j ($j = 0, \dots, N^{(1)}$) be the orthogonal matrix written by

$$V_j = \begin{bmatrix} |\mathbf{v}_j^{(1)}\rangle \otimes |\mathbf{v}_0^{(2)}\rangle \otimes \dots \otimes |\mathbf{v}_0^{(d-1)}\rangle \otimes |\mathbf{v}_0^{(d)}\rangle, \dots, \\ |\mathbf{v}_j^{(1)}\rangle \otimes |\mathbf{v}_0^{(2)}\rangle \otimes \dots \otimes |\mathbf{v}_0^{(d-1)}\rangle \otimes |\mathbf{v}_{N^{(d)}}^{(d)}\rangle, \\ |\mathbf{v}_j^{(1)}\rangle \otimes |\mathbf{v}_0^{(2)}\rangle \otimes \dots \otimes |\mathbf{v}_1^{(d-1)}\rangle \otimes |\mathbf{v}_0^{(d)}\rangle, \dots, \\ |\mathbf{v}_j^{(1)}\rangle \otimes |\mathbf{v}_0^{(2)}\rangle \otimes \dots \otimes |\mathbf{v}_1^{(d-1)}\rangle \otimes |\mathbf{v}_{N^{(d)}}^{(d)}\rangle, \dots, \\ |\mathbf{v}_j^{(1)}\rangle \otimes |\mathbf{v}_{N^{(2)}}^{(2)}\rangle \otimes \dots \otimes |\mathbf{v}_{N^{(d-1)}}^{(d-1)}\rangle \otimes |\mathbf{v}_{N^{(d)}}^{(d)}\rangle \end{bmatrix}$$

and V be

$$V = [V_0, V_1, \dots, V_{N^{(1)}}].$$

Since V and TV are orthogonal matrices, we have $V^TV = I$ and the following another description:

$$\begin{aligned}\delta_{\mathbf{j},\mathbf{k}} &= \sum_{\mathbf{l}=0}^{[N^{(1)}, N^{(2)}, \dots, N^{(d)}]} \prod_{m=1}^d \overline{\left(v_{l^{(m)}}^{(m)}(p_{l^{(m)}}^{(m)}(j^{(m)}))\right)} \left(v_{l^{(m)}}^{(m)}(p_{l^{(m)}}^{(m)}(k^{(m)}))\right) \\ &= \sum_{\mathbf{l}=0}^{[N^{(1)}, N^{(2)}, \dots, N^{(d)}]} \prod_{m=1}^d \overline{(p_{l^{(m)}}^{(m)}(j^{(m)}))} (p_{l^{(m)}}^{(m)}(k^{(m)})) \overline{v_{l^{(m)}}^{(m)}} v_{l^{(m)}}^{(m)} \\ &= \sum_{\mathbf{l}=0}^{[N^{(1)}, N^{(2)}, \dots, N^{(d)}]} \prod_{m=1}^d \overline{(p_{l^{(m)}}^{(m)}(j^{(m)}))} (p_{l^{(m)}}^{(m)}(k^{(m)})) \mu^{(m)}(\lambda_{l^{(m)}}^{(m)}),\end{aligned}\quad (5)$$

where $\mathbf{j} = [j^{(1)}, j^{(2)}, \dots, j^{(d)}]$, $\mathbf{k} = [k^{(1)}, k^{(2)}, \dots, k^{(d)}]$ and $\mathbf{l} = [l^{(1)}, l^{(2)}, \dots, l^{(d)}]$ with

$$\delta_{\mathbf{j},\mathbf{k}} = \begin{cases} 1 & (\mathbf{j} = \mathbf{k}), \\ 0 & (\mathbf{j} \neq \mathbf{k}). \end{cases}$$

Equation (5) means that $\mu^{(m)}$ can be regarded as the weight function of the orthogonal polynomials $p_{l^{(m)}}^{(m)}$, similar to 2-dimensional case.

2.2. CTQW

This section is devoted to the definition of CTQW related to J_n of birth and death chains.

Let $Y^{(l)}: \mathbb{R} \rightarrow \{0, \dots, N^{(l)}\}$ be the random variable about the position of the quantum walker of CTQW, where $l \in \{1, \dots, d\}$. We consider the following state space:

$$\mathcal{H} = \text{span}\{|x^{(1)}\rangle \otimes \dots \otimes |x^{(d)}\rangle : x^{(j)} \in \{0, \dots, N^{(j)}\}, j = 1, \dots, d\}.$$

Let $\Psi_t \in \mathcal{H}$ be the state of the quantum walker at time $t \in \mathbb{R}_{\geq 0}$. We define the time evolution operator U of time Δt as follows:

$$\begin{aligned}U(\Delta t) &= e^{i\Delta t J_N} \\ &= \exp \left[i\Delta t \sum_{\mathbf{j}=0}^{[N^{(1)}, N^{(2)}, \dots, N^{(d)}]} \left\{ \left(\sum_{k=1}^d q^{(k)} \lambda_{j^{(k)}}^{(k)} \right) \left(\bigotimes_{l=1}^d |\mathbf{v}_{j^{(l)}}^{(l)}\rangle \right) \left(\bigotimes_{l=1}^d \langle \mathbf{v}_{j^{(l)}}^{(l)}| \right) \right\} \right] \\ &= \sum_{\mathbf{j}=0}^{[N^{(1)}, N^{(2)}, \dots, N^{(d)}]} \left[\exp \left\{ i\Delta t \left(\sum_{k=1}^d q^{(k)} \lambda_{j^{(k)}}^{(k)} \right) \right\} \left(\bigotimes_{l=1}^d |\mathbf{v}_{j^{(l)}}^{(l)}\rangle \right) \left(\bigotimes_{l=1}^d \langle \mathbf{v}_{j^{(l)}}^{(l)}| \right) \right],\end{aligned}\quad (6)$$

where $\mathbf{j} = \{j^{(1)}, \dots, j^{(d)}\}$. Then the time evolution of CTQW is written by

$$\Psi_{t+\Delta t} = U(\Delta t)\Psi_t$$

and the transition probability is given by

$$\mathbb{P}(\mathbf{Y}(t + \Delta t) = \mathbf{k} \mid \mathbf{Y}(t) = \mathbf{j}) = \left| \left(\bigotimes_{l=1}^d \langle k^{(l)} | \right) U(\Delta t) \left(\bigotimes_{l=1}^d | j^{(l)} \rangle \right) \right|^2, \quad (7)$$

where $\mathbf{Y}(t) = (Y^{(1)}(t), \dots, Y^{(d)}(t))$, $\mathbf{j} = (j^{(1)}, \dots, j^{(d)})$, $\mathbf{k} = (k^{(1)}, \dots, k^{(d)})$, $j^{(l)} \in \{0, \dots, N^{(l)}\}$ and $k^{(l)} \in \{0, \dots, N^{(l)}\}$ with $l = 1, \dots, d$. We put the transition probability of l th dimension defined by

$$\mathbb{P}^{(l)}(Y^{(l)}(t + \Delta t) = k^{(l)} \mid Y^{(l)}(t) = j^{(l)}) = |\langle k^{(l)} | \exp(i\Delta t J_{N^{(l)}}) | j^{(l)} \rangle|^2.$$

3. Results

This section is devoted to our main results.

Theorem 1. *The transition probability $\mathbb{P}(\mathbf{Y}(t + \Delta t) = \mathbf{k} \mid \mathbf{Y}(t) = \mathbf{j})$ satisfies the following equation:*

$$\mathbb{P}(\mathbf{Y}(t + \Delta t) = \mathbf{k} \mid \mathbf{Y}(t) = \mathbf{j}) = \prod_{l=1}^d \mathbb{P}^{(l)}(Y^{(l)}(t + q^{(l)} \Delta t) = k^{(l)} \mid Y^{(l)}(t) = j^{(l)}). \quad (8)$$

Proof. [Proof of Theorem 1] By direct calculation, we have

$$\begin{aligned} & \mathbb{P}^{(l)}(Y^{(l)}(t + \Delta t) = k^{(l)} \mid Y^{(l)}(t) = j^{(l)}) \\ &= \left| \langle k^{(l)} \left| \sum_{m=0}^{N^{(l)}} \left\{ \exp(i\Delta t \lambda_m^{(l)}) |\mathbf{v}_m^{(l)}\rangle \langle \mathbf{v}_m^{(l)}| \right\} \right| j^{(l)} \rangle \right|^2 \\ &= \left| \sum_{m=0}^{N^{(l)}} \left\{ \exp(i\Delta t \lambda_m^{(l)}) (\langle k^{(l)} | \mathbf{v}_m^{(l)} \rangle \langle \mathbf{v}_m^{(l)} | j^{(l)} \rangle) \right\} \right|^2 \\ &= \left| \sum_{m=0}^{N^{(l)}} \left\{ \exp(i\Delta t \lambda_m^{(l)}) v_m^{(l)}(p_m^{(l)}(k^{(l)})) \overline{v_m^{(l)}(p_m^{(l)}(j^{(l)}))} \right\} \right|^2 \\ &= \left| \sum_{m=0}^{N^{(l)}} \left\{ \exp(i\Delta t \lambda_m^{(l)}) (p_m^{(l)}(k^{(l)}) \overline{p_m^{(l)}(j^{(l)})}) \mu^{(l)}(\lambda_m^{(l)}) \right\} \right|^2. \end{aligned} \quad (9)$$

By substituting Eqs. (6) and (9) into Eq. (7), we obtain

$$\begin{aligned} & \mathbb{P}(\mathbf{Y}(t + \Delta t) = \mathbf{k} \mid \mathbf{Y}(t) = \mathbf{j}) \\ &= \left| \sum_{\mathbf{m}=0}^{[N^{(1)}, N^{(2)}, \dots, N^{(d)}]} \left[\exp \left\{ i\Delta t \left(\sum_{h=1}^d q^{(h)} \lambda_{m^{(h)}}^{(h)} \right) \right\} \left(\prod_{l=1}^d \langle k^{(l)} | \mathbf{v}_{m^{(l)}}^{(l)} \rangle \langle \mathbf{v}_{m^{(l)}}^{(l)} | j^{(l)} \rangle \right) \right] \right|^2 \\ &= \left| \sum_{m^{(1)}=0}^{N^{(1)}} \sum_{\mathbf{m}=[m^{(1)}, 0, \dots, 0]}^{[m^{(1)}, N^{(2)}, \dots, N^{(d)}]} \left[\exp \left\{ i\Delta t \left(\sum_{h=1}^d q^{(h)} \lambda_{m^{(h)}}^{(h)} \right) \right\} \right] \right|^2 \end{aligned}$$

$$\begin{aligned}
& \times \left(\prod_{l=1}^d \langle k^{(l)} | \mathbf{v}_{m^{(l)}}^{(l)} \rangle \langle \mathbf{v}_{m^{(l)}}^{(l)} | j^{(l)} \rangle \right) \Bigg]^2 \\
& = \left| \sum_{m^{(1)}=0}^{N^{(1)}} \left\{ \exp \left(i q^{(1)} \Delta t \lambda_{m^{(1)}}^{(1)} \right) \langle k^{(1)} | \mathbf{v}_{m^{(1)}}^{(1)} \rangle \langle \mathbf{v}_{m^{(1)}}^{(1)} | j^{(1)} \rangle \right\} \right|^2 \\
& \quad \times \left| \sum_{\mathbf{m}=[*,0,\dots,0]}^{[*,N^{(2)},\dots,N^{(d)}]} \left[\exp \left\{ i \Delta t \left(\sum_{h=2}^d q^{(h)} \lambda_{m^{(h)}}^{(h)} \right) \right\} \left(\prod_{l=2}^d \langle k^{(l)} | \mathbf{v}_{m^{(l)}}^{(l)} \rangle \langle \mathbf{v}_{m^{(l)}}^{(l)} | j^{(l)} \rangle \right) \right] \right|^2 \\
& = \mathbb{P}^{(1)}(Y^{(1)}(t + q^{(1)} \Delta t) = k^{(1)} \mid Y^{(1)}(t) = j^{(1)}) \\
& \quad \times \left| \sum_{\mathbf{m}=[*,0,\dots,0]}^{[*,N^{(2)},\dots,N^{(d)}]} \left[\exp \left\{ i \Delta t \left(\sum_{h=2}^d q^{(h)} \lambda_{m^{(h)}}^{(h)} \right) \right\} \left(\prod_{l=2}^d \langle k^{(l)} | \mathbf{v}_{m^{(l)}}^{(l)} \rangle \langle \mathbf{v}_{m^{(l)}}^{(l)} | j^{(l)} \rangle \right) \right] \right|^2 \\
& = \prod_{l=1}^d \mathbb{P}^{(l)}(Y^{(l)}(t + q^{(l)} \Delta t) = k^{(l)} \mid Y^{(l)}(t) = j^{(l)}).
\end{aligned}$$

This equation means that the transition probability of multi-dimensional CTQW can be calculated by the product of the transition probability of 1-dimensional CTQW. $\square$

From now on, we assume $N^{(l)} = N/d$ ($l = 1, \dots, d$) and $q^{(l)} = 1/d$ ($l = 1, \dots, d$). Let $\mathbf{j} = [j^{(1)}, \dots, j^{(d)}]$ be the initial position of the quantum walker and

$$\mathbf{Y}_{\mathbf{j}}(t) = [Y_{j^{(1)}}^{(1)}(t), \dots, Y_{j^{(d)}}^{(d)}(t)]$$

be the position at time $t \in \mathbb{R}$. Each $Y_{j^{(l)}}^{(l)} \in \{0, \dots, N/d\}$ can be regarded as the random variable. By Eq. (8), the transition probability is written as follows:

$$\begin{aligned}
\mathbb{P}(\mathbf{Y}_{\mathbf{j}}(t) = \mathbf{k}) &= \prod_{l=1}^d \mathbb{P}^{(l)}(Y_{j^{(l)}}^{(l)}(q^{(l)}t) = k^{(l)}) \\
&= \prod_{l=1}^d \mathbb{P}^{(l)}\left(Y_{j^{(l)}}^{(l)}\left(\frac{t}{d}\right) = k^{(l)}\right). \tag{10}
\end{aligned}$$

When t of left-hand side of Eq. (10) is scaled to t/d , this equation means that $Y_{j^{(l)}}^{(l)}$, ($l = 1, \dots, d$) are independent. When we assume $N^{(l)} = 1$ ($l = 1, \dots, d = N$), we can regard CTQW on the path as the tensor product by N th power of CTQW on the edge. Thus we have the following proposition.

Proposition 1. *The random variable about the position of CTQW related to the 1-dimensional Ehrenfest model $Y_0 = Y_0^{(1)} + Y_0^{(2)} + \dots + Y_0^{(d)}$ obeys the binomial distribution $B(d, \sin^2 \frac{t}{d})$.*

Proof. [Proof of Proposition 1] For each $l = 0, \dots, d$, we have

$$J_{N^{(l)}} = J_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then we obtain

$$\begin{aligned} e^{itJ_{N^{(l)}}} &= \sum_{k=0}^{\infty} \frac{(it)^k}{k!} J_{N^{(l)}}^k \\ &= \sum_{k=0}^{\infty} \frac{(it)^{2k}}{(2k)!} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \sum_{k=0}^{\infty} \frac{(it)^{2k+1}}{(2k+1)!} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \cos t & i \sin t \\ i \sin t & \cos t \end{bmatrix}. \end{aligned}$$

Then we get $\mathbb{P}^{(l)}(Y_0^{(l)}(t) = 1) = \sin^2 t$. By Eq. (10), we have

$$\begin{aligned} \mathbb{P}\left(Y_0(t) = \sum_{l=1}^d j^{(l)}\right) &= \mathbb{P}(\mathbf{Y}_0(t) = \mathbf{j}) \\ &= \prod_{l=1}^d \mathbb{P}^{(l)}\left(Y_0^{(l)}\left(\frac{t}{d}\right) = j^{(l)}\right), \end{aligned}$$

where $\mathbf{j} = [j^{(1)}, \dots, j^{(d)}]$, $j^{(l)} \in \{0, 1\}$ ($l = 1, \dots, d$). Thus we obtain the conclusion. $\square$

This is another easier calculation method for the analysis of the 1-dimensional CTQW related to Ehrenfest model, obtained in Ref. 16.

Next, we consider the random variable $Z_{\mathbf{j}}^{(d)}$ of position at time $T \in \mathbb{R}$, defined by

$$Z_{\mathbf{j}}^{(d)}(T) = \frac{1}{d} \{Y_{j^{(1)}}^{(1)}(T) + \dots + Y_{j^{(d)}}^{(d)}(T)\}.$$

Theorem 2. For any constant $T \geq 0$, we have

$$\frac{Z_{\mathbf{j}}^{(d)}(T) - \mathbb{E}(Y_{j^{(1)}}^{(1)}(T))}{\sqrt{\frac{V(Y_{j^{(1)}}^{(1)}(T))}{d}}} \Rightarrow N(0, 1) \quad (d \rightarrow \infty), \quad (11)$$

where $\Rightarrow$ stands for the weak convergence and $N(0, 1)$ is the standard Gaussian distribution.

Proof. [Proof of Theorem 2] Noting that $Y_{j^{(1)}}^{(1)}(T), \dots, Y_{j^{(d)}}^{(d)}(T)$ are independent and identically distributed. Since T is a constant, variance of $Y_{j^{(1)}}^{(1)}(T), \dots, Y_{j^{(d)}}^{(d)}(T)$ are finite. Then we have Eq. (11) by central limit theorem. $\square$

4. Summary

In this paper, we considered multi-dimensional birth and death chains and CTQW related to them. We obtained the time scaled independence between multiple dimensions about the transition probability of CTQW. After that, we limited our discussion to the symmetric multi-dimensional Ehrenfest model case. We got the different proof from the previous study to show that the random variable about the position of CTQW related to the 1-dimensional Ehrenfest model obeys the binomial distribution. We also had a random variable which is related to our models and converges to the standard Gaussian distribution. Considering CTQW on more general graph and analyzing them using the dividing method of Proposition 1 is one of the interesting future problems.

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